

Divya Nallawar

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Data Scientist | Time-Series Analysis, Experimentation & Anomaly Detection

EDUCATION

University of Colorado (CUB), Boulder, CO | master's in data science (GPA: 3.91)

Aug 2023 - May 2025

Savitribai Phule Pune University, India | Bachelor's in Electronics and telecommunication (GPA: 8.45)

Aug 2017 - May 2021

WORK EXPERIENCE

University of Colorado, Boulder

Boulder, CO

Research Assistant (Data Science & Time-Series Modeling)

Apr 2024 – Present

- Developed and benchmarked forecasting models (SARIMA, LSTM, Transformer-based models) on WiFi time-series data across **144 campus buildings**, improved **forecast accuracy** to support capacity planning decisions **by 20–35%** compared to statistical baselines.
- Conducted comparative model analysis across **5+ forecasting approaches** to evaluate accuracy, error variance, stability, and inference efficiency for large-scale deployment.
- Engineered temporal behavioral features (inter-arrival time, flow duration, throughput) across **100+ IP flows** to distinguish human vs automated traffic patterns.
- Designed reproducible Python-based preprocessing pipelines to transform **1M+ packet captures** into clean, analysis-ready datasets, reducing preprocessing time by **70%**.
- Developed and integrated software for a **\$5M** project on securing cellular communications. Developed the operations (Ops) module for the system and participated in integration of an end-to-end system prototype.
- Built analytical dashboards (Python, Tableau) to visualize trends, anomalies, and forecasting outputs for operational decision support.

StateSimplify LLC

Boulder, CO

Data Science Intern

Jun 2024 – May 2025

- Designed and deployed an end-to-end ML pipeline on **100k+ federal contract records**, **reducing prediction error by 93%** relative to baseline models.
- Built NLP classification pipelines using BERT and RoBERTa embeddings to analyze unstructured contract descriptions, improving trend **detection accuracy by 30–40%**.
- Engineered regression and ensemble models (Linear Regression, Random Forest, LSTM, BERT-Regression) optimized via GridSearchCV.
- Evaluated model robustness using RMSE, MAE, R^2 , k-fold cross-validation, and variance analysis to ensure reproducibility and stability.
- Built a scalable mental health chatbot leveraging OpenAI APIs and biomedical datasets to deliver real-time evidence-based responses with **95% accuracy**.

Amdocs

Pune, India

Software Developer

Aug 2021 - Jul 2023

- Developed data-driven automation systems using Python and SQL, improving operational **efficiency by 80%** and **reducing manual intervention by 98%**.
- Delivered 12 end-to-end automation solutions, **reducing manual errors by 60%** and improving workflow reliability.
- Integrated Azure-based deployment pipelines enabling zero-downtime updates and **30% performance improvements**.

SKILL SET

Programming: Python, SQL, R

Data Analysis: EDA, Statistical Modeling, Hypothesis Testing, Experiment Design, A/B Testing

Machine Learning: Regression, Classification, Ensemble Models, Time-Series Forecasting, LSTM, Transformers, NLP, LLMs

Model Evaluation: RMSE, MAE, R^2 , Cross-Validation, Drift Detection

Libraries & Tools: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Statsmodels, Seaborn

Visualization: Tableau, Power BI, Matplotlib, Streamlit

Cloud & Deployment: GCP, Azure, Docker, FastAPI

Databases: MySQL, PostgreSQL, MongoDB, Elasticsearch

PROJECTS

Prediction of Potentially Hazardous Asteroids | Python, A/B testing, ANOVA, EDA, Random Forest, ANN

Built Random Forest and ANN classification models achieving 97.8% prediction accuracy. Applied ANOVA and A/B testing for validations

Time Series & Dimensionality Reduction Study | Python, R, SVD, NMF, SSA, Data Reconstruction

Conducted comparative analysis of SVD and NMF on time-series and image data, achieving 71% lower reconstruction MSE using SVD. Applied Singular Spectrum Analysis (SSA) for trend extraction and signal decomposition.

Wi-Fi Time Series Analytics Platform | Python, Pandas, Plotly, Statsmodels, FastAPI, Angular, PostgreSQL

Built a full-stack time-series analytics platform leveraging SQL-based retrieval and preprocessing; implemented STL decomposition, FFT spectral reconstruction (top-N), and PELT changepoint detection via FastAPI and Angular for interactive modeling and visualization.

PUBLICATIONS

Comparing and Contrasting Time series Prediction/Forecast Methods on Campus Traffic Data (manuscript under review)